* INT 21h -> Stops our program and takes Input/Output from the user;

* The value stored in AH determines the operation
* AH (1) -> Single character input saved in AL
* AH (2) -> Single character output saved in DL
* AH (9) -> String output

* converts the input to ascii and store
* So, for number input we should immediately do SUB AL, 30h to match the input number
* output is the ascii equivalent character
* Similarly, immediately do ADD AL, 30h to match the output number

* any variable's starting address is stored in DS. for example,
temp DB "Hello World$" 's starting address is saved in Data Segment

* LEA (Load Effective Address) -> Starting address to Ending Address er offset.
* LEA is used with DX register

LEA DX, variable

*** For, multiple input, for example for input 47,

4 * 10
7
40 + 7

* Again for 536,

5 * 100
3 * 10
6
500 + 30 + 6

* To show output, for example 47,
47/10 = quotient 4 in AL, remainder 7 in AH

* 0A -> newl -> new line (under the cursor)
* 0D -> cret -> carriage return (start of the line)

* Flag stores some certain conditions or state of the processor after an arithmetic operation
* Carry Flag (CF) -> 1 if there is a carry out from the MSB during addition or a borrow into the MSB during subtraction

* Parity Flag (PF) -> 1 if the low byte of a result of an operation contains an even number of ones

* Auxiliary Flag (AF) -> 1 when there is a carry out from bit number 3 to bit number 4 in

addition or borrow into bit number 3 during subtraction. (Note that the bit number starts from 0)

* Zero Flag (ZF) -> 1 if the result of an operation in 0
* Sign Flag (SF) -> 1 if the result of an operation in 0
* Overflow Flag (OF) -> 1 if there is an overflow